Project Synopsis

Project Title

SMS Spam Classifier Using Machine Learning

Objective

The objective of this project is to develop a machine learning-based application that can automatically classify SMS messages as **Spam** or **Ham (Not Spam)**. The system is designed to help users quickly identify unwanted or fraudulent messages through a simple and user-friendly interface.

Technology Used

- Python
- Scikit-learn
- Streamlit
- NLTK
- Pandas
- NumPy
- Pickle

Dataset

The project uses the **SMS Spam Collection Dataset** obtained from the **UCI Machine Learning Repository/Kaggle**. The dataset contains labeled SMS messages that are used to train and evaluate the classification model.

Methodology

- Collect the SMS Spam Collection Dataset.
- Perform text preprocessing, including lowercase conversion, punctuation removal, stopword removal, and stemming.
- Convert the cleaned text into numerical features using the **TF-IDF Vectorizer**.
- Train the classification model using the **Multinomial Naive Bayes** algorithm.
- Save the trained model and vectorizer using Pickle.
- Develop a **Streamlit** web application to classify user-entered SMS messages.

Expected Outcome

The final outcome of this project is a web-based application that accurately predicts whether an SMS message is **Spam** or **Ham**. The application provides quick predictions through a simple and interactive user interface.

Future Scope

- Email spam detection.
- Multilingual spam classification.
- Implementation of deep learning models such as **LSTM** and **BERT**.
- Integration with real-time messaging platforms and APIs for automatic spam detection.